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1 PROFESSIONAL SUMMARY

Systems-oriented Engineering Leader & Venture Founder with a 12+ year track record of scaling global technical organizations from 0 to 1 across Asia, Australia, and the Americas. Expert in transforming legacy technical debt into high-availability enterprise stacks and bridging the gap between **MIT-tier research and commercial viability**. A strategic architect of **socially responsible technology**, I have led the development of complex platforms for **renewable energy, carbon tracking, and urban mobility**. Proven success in securing **\$250k+ in non-dilutive innovation funding** and building high-retention, cross-functional cultures that align engineering excellence with meaningful environmental and social impact.

2 PROFESSIONAL EXPERIENCE

Pulse Clinic Singapore & Bangkok, Thailand
Chief Technology Officer (CTO) 2025 – Now

C#, TypeScript, Docker, Agile, CD/CI, Terraform, GitHub Actions, Jira

- **Engineering Transformation:** Re-engineered a decade of legacy technical debt by implementing CI/CD, Agile methodologies, and Clean Code principles.
- **Organizational Growth:** Built a cross-functional Agile team (UX/Product/Eng) by vetting hundreds of candidates to transform a developer-only silo.
- **System Reliability:** Improved platform uptime by over 90%, reducing critical outages from 3x weekly to <1x monthly.
- **Regional Leadership:** Strategized technology for 200+ employees serving tens of thousands of customers across 6 international markets (TH, MY, SG, PH, HK).
- **Digital Transformation:** Architected and executed migrating from a legacy in-house Clinic Management System to an enterprise solution with zero data incidents.

Flo Energy Singapore & Melbourne, Australia
Tech Lead -> Engineering Manager 2021 – 2025

Kotlin, TypeScript, GraphQL, Kubernetes, Docker, Agile, CD/CI, Terraform, Bitbucket, Jira, IoT

- **Organizational Scaling:** Built 4 global engineering teams from zero, scaling from 4 to 20+ members in 1.5 years with established onboarding and innovation culture.
- **Product Expansion:** Oversaw 0-to-1 product development and expansion into international markets, including Renewable Energy Certificate implementations.
- **Strategic Integration:** Led high-stakes technical integrations with Singapore's Energy Market Company and drove evidence-based corporate decision-making.
- **Operational Excellence:** Hardened infrastructure by modernizing CI/CD pipelines, automated testing, and global privacy/security policies.
- **Technical Strategy:** Directed quantitative research and corporate deliberation to steer high-level technology improvements and IoT architecture.

Singapore - MIT Alliance for Research and Technology (SMART)

Singapore
2020 - 2021

Tech Lead Software Engineer

Node.js, Angular, Express, TypeScript, WSL, Linux, Ubuntu, ECharts, Open Layers, Python

- **System Modernization:** Architected and engineered a system-wide UI overhaul for SimMobility's agent-based simulation platform, migrating legacy components to a high-performance Angular framework.

Michoacan University of Saint Nicholas of Hidalgo (UMSNH)

Morelia, Mexico
2020 - 2021

Lecturer - Urban Data

ArcGIS, QGIS, CARTO, Open Data

- **Curriculum Design:** Developed the original syllabus and technical materials for a graduate-level Urban Data course from the ground up.
- **Geospatial Instruction:** Taught Open Data philosophy and practical urban planning applications using QGIS and CARTO mapping suites.

Massachusetts Institute of Technology (MIT) + SMART

Boston, USA & Singapore
2014 - 2020

Project Lead Software Engineer

Android, Java, JavaScript, Mapbox API, PostgreSQL, REST, Spring, Jersey, Google Maps API, PostGIS, R, R Studio

- **Team Leadership:** Led a team of 9 in the R&D of Flocktracker, a specialized platform for high-precision, georeferenced field surveys.
- **Full-Stack Architecture:** Architected and programmed the end-to-end cloud stack, including the RESTful backend and the Android client-side UX.
- **Engineered Resilience:** Developed a custom asynchronous system for geospatial data collection and route tracing, optimized for unreliable offline environments.
- **Global Scale:** Directed community outreach and negotiated software deployments in 30+ locations across 4 continents.
- **Technology Commercialization:** Successfully translated research into a commercial venture by launching Fuente to take Flocktracker to market.

Kota Kita

Surakarta, Indonesia

Technologist

2014 - 2014

Mapbox, Database Design, Web Map Design

- **Civic Technology:** Designed and developed a participatory process database and technical infrastructure for urban engagement.
- **Geospatial Visualization:** Engineered front-end web map designs to visualize community-driven data.

Urban Launchpad + New Market Goods

San Francisco, USA & Dhaka, Bangladesh

Technologist

2014 - 2014

Android, Mapbox, Web Dev, Kickstarter

- **Technical Strategy:** Acted as strategy consultant and lead developer for a Bay Area social startup, overseeing front-end web and Android programming.
- **Crowdsourcing launch & Data Transparency:** Designed a transparency database and produced all technical materials for a successful Kickstarter crowdfunding campaign.

MIT + Urban Travel Logistics

Mexico City, Mexico

Software Engineer

2013 – 2013

Android, System Research

- **Mobile Research Systems:** Developed and modified specialized Android applications to conduct system research on urban transport logistics.
- **Operations & Scholarship:** Managed large-scale data collection operations and contributed to academic scholarship regarding software research methodologies.

Engineering Institute, National Autonomous University of Mexico

Mexico City, Mexico

2011 – 2014

Research Assistant

FORTRAN, AutoLISP, AutoCAD, Finite Element Modeling, Discrete Element Modeling

- **Geotechnical Modeling:** Developed 3D CAD tools for finite and discrete element modeling of soil dams and shafts under seismic loads using FORTRAN and AutoLISP.

3 ENTREPRENEURSHIP

Fuente Tech

Singapore

Founding Chief Executive Officer (CEO),

2020 – 2026

Tech Lead

Android, Java, Gradle, GitHub Actions, Docker, AWS, Mapbox, PostgreSQL, Spring, Jersey

- **Capital Acquisition:** Raised \$200,000+ USD in non-dilutive funding from the National Science Foundation (NSF), MIT, and SMART.
- **IP Strategy:** Negotiated complex technology licensing agreements for the Flocktracker platform with MIT and the National University of Singapore (NUS).
- **System Hardening:** Architected and implemented thousands of unit and integration tests to ensure high availability for enterprise-grade deployments.
- **Cloud Infrastructure:** Orchestrated the migration of the Flocktracker stack to AWS to transition the platform from research to commercial scale.

M+B

Singapore

Founding Chief Technology Officer (CTO)

2020 – 2021

WordPress, WPML, Stripe, WooCommerce

- **Platform Engineering:** Designed and implemented a multilingual publication platform using WordPress and WPML.
- **E-commerce Automation:** Built a custom web store to fully automate the sales lifecycle, from end-user purchase through to fulfillment and accounting.

Joro (now Commons Earth)

Boston & Oakland, USA

Founding Software Advisor

2017 – 2020

Node.js, React, React Native, MongoDB

- **Technical Foundation:** Designed database and full-stack architecture for a React Native carbon-footprinting platform, incorporating AI-driven UX features.
- **Strategic Advisory:** Guided software architecture, documentation standards, and engineering hiring to scale the technical organization.

4 EDUCATION

**Massachusetts Institute of Technology
(MIT)**

Boston, USA
2016–2019

Master of City Planning

Urban data, mobility and sustainability

Autonomous Metropolitan University (UAM)

Mexico City, Mexico

2011–2013

Studies in Social Anthropology

Urban conflict

**National Autonomous University of Mexico
(UNAM)**

Mexico City, Mexico

2009–2012

Civil Engineer

Soil and seismic engineering